

Dhcp - GNS3

File Edit View Control Node Annotate Tools Help

All devices

Filter

- ATM switch
- cisco7200
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- NAT
- VPCS

+ New template

Enterprise Architecture SBI-BANK

DHCP-server

R1

ip dhcp pool SBI-BANK
network 192.168.1.0 255.255.255.0
default router 192.168.1.100
dns-server 8.8.8.8 1.1.1.1

Switch1

PC1

PC2

IP Address
Subnet mask
Gateway
DNS

Topology Summary

Node	Console
PC1	telnet localhost:5002
PC2	telnet localhost:5004
R1	telnet localhost:5000
Switch1	none

Servers Summary

Sowmith CPU 20.4%, RAM 62.9%

Console

GNS3 management console.
Running GNS3 version 2.2.61 on Windows (64-bit) with Python 3.13.14 Qt 6.10.0 and PyQt 6.10.2.
Copyright (c) 2006-2026 GNS3 Technologies.
Use Help -> GNS3 Doctor to detect common issues.

=>

- ✓ Dynamic Host Configuration Protocol (Discover)

0000	ff ff ff ff ff ff 00 50	79 66 68 00 08 00 45 10	P yfh...E..
0010	01 88 00 00 00 00 10 11	a9 56 00 00 00 00 ff ff	V.....
0020	ff ff 00 44 00 43 01 74	59 f7 01 01 06 00 d4 48	...D-C-t Y.....H
0030	98 1d 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0040	00 00 00 00 00 00 00 50	79 66 68 00 00 00 00 00	P yfh...E..
0050	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0060	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
0070	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	

Client MAC = AA

> Frame 10: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
 > Ethernet II, Src: ca:01:c3:f8:00:70 (ca:01:c3:f8:00:70), Dst: Private_66:68:00 (00:50:79:66:68:00)
 > Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.1
 > User Datagram Protocol, Src Port: 67, Dst Port: 68
 ✓ Dynamic Host Configuration Protocol (Offer)

Message type: Boot Reply (2)
 Hardware type: Ethernet (0x01)
 Hardware address length: 6
 Hops: 0
 Transaction ID: 0xd448981d
 Seconds elapsed: 0
 > Bootp flags: 0x0000 (Unicast)
 Client IP address: 0.0.0.0
 Your (client) IP address: 192.168.1.1
 Next server IP address: 0.0.0.0
 Relay agent IP address: 0.0.0.0
 Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Client hardware address padding: 00000000000000000000
 Server host name not given
 Boot file name not given
 Magic cookie: DHCP
 > Option: (53) DHCP Message Type (Offer)
 > Option: (54) DHCP Server Identifier (192.168.1.100)
 > Option: (51) IP Address Lease Time
 > Option: (58) Renewal Time Value
 > Option: (59) Rebinding Time Value
 > Option: (1) Subnet Mask (255.255.255.0)
 > Option: (3) Router
 > Option: (6) Domain Name Server
 > Option: (255) End
 Padding: 00000000000000000000

DHCP Server will Respond with OFFER Message

SMAC RR1 DMAC FF:FF:FF:FF

SIP 192.168.1.100 DIP 255.255.255

UDP,SP=67,DP=68

DHCP Payload

Type of DHCP Packet(53)=OFFER

Client MAC = AA ,your IP=192.168.1.1,subnetmark(1)=255.255.255.0

Default router(3)=192.168.1.100, DNS(6)=8.8.8.8 1.1.1.1,

DHCP-Server ID(54)=192.168.1.100, lease time(51)=1 day

```
0000 00 50 79 66 68 00 ca 01 c3 f8 00 70 08 00 45 00  ·Pyfh··· ··p·E·
0010 01 48 00 00 00 00 ff 11 36 ef c0 a8 01 64 c0 a8  ·H····· 6···d·
0020 01 01 00 43 00 44 01 34 91 fc 02 01 06 00 d4 48  ···C·D·4 ·····H
0030 98 1d 00 00 00 00 00 00 00 00 c0 a8 01 01 00 00  ······ ······
0040 00 00 00 00 00 00 00 50 79 66 68 00 00 00 00  ······P yfh····
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ······ ······
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ······ ······
```

No.: 10 · Time: 98.984355 · Source: 192.168.1.100 · Destination: 192.168.1.1 · Protocol: DHCP · Length: 342 · Info: DHCP Offer - Transaction ID 0xd448981d

☒ Show packet bytes Layout: Vertical (Stacked) ▾

```
PC1 will generate Request message
SMAC AA          DMAC FF:FF:FF:FF
SIP 0.0.0.0      DIP 255.255.255.255
UDP,SP=68,DP=67
DHCP Payload
Type of DHCP Packet(53)=Request
Client MAC = AA,DHCP Server ID(54)=192,168.1.100,
Requested ID=192.168.1.1
```

0000	ca 01 c3 f8 00 70	00 50 79	66 68 00 08 00 45 10		p P yfh	...E
0010	01 88 00 00 00 10 11	a9 56 00 00 00 00 ff ff				V
0020	ff ff 00 44 00 43 01 74	84 60 01 01 06 00 d4 48			D C t	...H
0030	98 1d 00 00 00 c0 a8	01 01 00 00 00 00 d4 00				
0040	00 00 00 00 00 00 50	79 66 68 00 00 00 00 00			P yfh	
0050	00 00 00 00 00 00 00	00 00 00 00 00 00 00 00				
0060	00 00 00 00 00 00 00	00 00 00 00 00 00 00 00				
0070	00 00 00 00 00 00 00	00 00 00 00 00 00 00 00				
0080	00 00 00 00 00 00 00	00 00 00 00 00 00 00 00				

> Frame 12: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
 > Ethernet II, Src: ca:01:c3:f8:00:70 (ca:01:c3:f8:00:70), Dst: Private_66:68:00 (00:50:79:66:68:00)
 > Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.1
 > User Datagram Protocol, Src Port: 67, Dst Port: 68
 > Dynamic Host Configuration Protocol (ACK)

Message type: Boot Reply (2)
 Hardware type: Ethernet (0x01)
 Hardware address length: 6
 Hops: 0
 Transaction ID: 0xd448981d
 Seconds elapsed: 0
 > Bootp flags: 0x0000 (Unicast)
 Client IP address: 192.168.1.1
 Your (client) IP address: 192.168.1.1
 Next server IP address: 0.0.0.0
 Relay agent IP address: 0.0.0.0
 Client MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Client hardware address padding: 00000000000000000000
 Server host name not given
 Boot file name not given
 Magic cookie: DHCP
 > Option: (53) DHCP Message Type (ACK)
 > Option: (54) DHCP Server Identifier (192.168.1.100)
 > Option: (51) IP Address Lease Time
 > Option: (58) Renewal Time Value
 > Option: (59) Rebinding Time Value
 > Option: (1) Subnet Mask (255.255.255.0)
 > Option: (3) Router
 > Option: (6) Domain Name Server
 > Option: (255) End
 Padding: 00000000000000000000

DHCP server will generate ACK message

SMAC RR1

DMAC FF:FF:FF:FF

SIP 192.168.1.100

DIP 255.255.255

UDP,SP=67,DP=68

DHCP Payload

Type of DHCP Packet(53)=ACK

Client MAC = AA ,your IP=192.168.1.1,subnetmask(1)=255.255.255.0

Default router(3)=192.168.1.100, DNS(6)=8.8.8.8 1.1.1.1,

DHCP-Server ID(54)=192.168.1.100, lease time(51)=1 day

0030	98 1d 00 00 00 00 c0 a8 01 01 c0 a8 01 01 00 00	P yfh.....
0040	00 00 00 00 00 00 00 50 79 66 68 00 00 00 00 00	
0050	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0060	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0070	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0080	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0090	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Seconds elapsed (dhcp.secs), 2 bytes

☒ Show packet bytes Layout: Vertical (Stacked) ▾